

Exercise set 2

Problem 1:

Let X be exponentially distributed with density function $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.

- a)^R Make a sketch of this density (e.g. for $\lambda = 1$)
- b) Compute the expectation and the variance of X as a function of λ .

The lifetime of a type of light bulbs follows an exponential distribution with expectation 10 000 hours.

- c) Calculate the probability that the lifetime of a light bulb is less than 10 000 hours.
Calculate the probability that the lifetime of a light bulb is more than 5 000 hours.
Calculate the probability that the lifetime of a light bulb is between 5 000 and 10 000 hours.
- d)^R Redo all the calculations from point c) using the R function `pexp`.

Problem 2:

The number of accidents at a certain road per year is assumed to be Poisson distributed with parameter $\lambda = 2$.

- a) Calculate the probability that there will be exactly 2 accidents during one year.
Calculate the probability that there will be at least 3 accidents during one year.
- b) Calculate the probability that there will be exactly 2 accidents during half a year.
- c) Calculate the probability that there will be at least 20 accidents during ten years
(Hint for the manual calculation: By the Central Limit Theorem a Poisson distribution with expectation larger than 15 can be approximated by a normal distribution with the same expectation and variance as the Poisson distribution).
- d) Calculate the probability that there will be more than 2 accidents on this road during a year when we know that there has been at least one accident.
- e)^R Calculate the above probabilities using the R functions `dpois` and `ppois`.

Problem 3:

Let X be a random variable that has the beta-distribution given by

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 \leq x \leq 1, \alpha > 0, \beta > 0$$

where

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$$

and for integers $\Gamma(\alpha) = (\alpha - 1)!$

- a)^R Write up the formula for $f(x)$ and make a sketch of $f(x)$ for each of the cases (i) $\alpha = \beta = 1$, (ii) $\alpha = 2, \beta = 1$ and (iii) $\alpha = 1, \beta = 2$.
- b)^R Use R to draw samples of different sizes (e.g. size 10, 100, 1000 and 10000) from each of the beta-distributions specified in a). Calculate the average of each of these samples and compare to the expectation which for the beta distribution is given by the formula $E(X) = \alpha/(\alpha + \beta)$.

Problem 4:

Let $U_i, i = 1, \dots, n$ be independent and identically uniformly distributed on $(0, 2)$.

- a) Find $E(U_i)$ and $\text{Var}(U_i)$.
Explain why

$$X = \frac{U_1 + \dots + U_n - n}{\sqrt{n\text{Var}(U_i)}}$$

approaches a standard normally distributed variable when n increases.

- b)^R Let n be 5, 20 and 100 and use as many replications in the simulation as you deem necessary to obtain a histogram that gives a good approximation to the distribution for X .

Problem 5:

Do exercise 12.8 in Rizzo. Also make a histogram of the data using different choices of bin-width.

Note that you may obtain the dataset by first installing the package `gss` (e.g. by running `install.packages("gss")`, (only needs to be done once)). Then the dataset is available in a vector `buffalo` after calling

```
library(gss)
data(buffalo)
```